

NT-C-011 | .NET Full Stack | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Programming & Full Stack
Target learners	Learners targeting Microsoft-stack web application roles.
Prerequisites	Basic logic and computer use; prior coding is helpful.
Approved duration	20-24 weeks / 240-300 guided hours.
Final outcome	Build, test and deploy a secure database-backed application with ASP.NET Core APIs and a modern user interface.
Assessment profile	Programming/Platform

Approved tools / platforms

C#, .NET, ASP.NET Core, Entity Framework Core, SQL, HTML, CSS, JavaScript, React or Angular, Git.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.	Do not treat copied code or syntax recall as proof of implementation skill.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-011 | .NET Full Stack | NT-C-011-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Syntax, control flow, methods and types; OOP, collections, LINQ, async and testing
Completion evidence	Working console programs; Tested C# solution
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 1 - C# foundations

Objective: Syntax, control flow, methods and types

Observable evidence: Working console programs

Concept	Approved knowledge statement	Application behaviour	Evidence
Syntax	The formal rules used to write valid instructions in a programming language. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when syntax is appropriate; apply it in a .NET Full Stack task; verify the result.	Working console programs
control flow	The order in which conditions, loops and branches execute. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when control flow is appropriate; apply it in a .NET Full Stack task; verify the result.	Working console programs
methods	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when methods is appropriate; apply it in a .NET Full Stack task; verify the result.	Working console programs
types	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when types is appropriate; apply it in a .NET Full Stack task; verify the result.	Working console programs

Module 2 - Professional C#

Objective: OOP, collections, LINQ, async and testing

Observable evidence: Tested C# solution

Concept	Approved knowledge statement	Application behaviour	Evidence
OOP	A design approach that organises behaviour and data around objects and clear responsibilities. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when oop is appropriate; apply it in a .NET Full Stack task; verify the result.	Tested C# solution
collections	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when collections is appropriate; apply it in a .NET Full Stack task; verify the result.	Tested C# solution
LINQ	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when linq is appropriate; apply it in a .NET Full Stack task; verify the result.	Tested C# solution
async	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when async is appropriate; apply it in a .NET Full Stack task; verify the result.	Tested C# solution
testing	Systematic verification that expected behaviour and important failure cases work correctly. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when testing is appropriate; apply it in a .NET Full Stack task; verify the result.	Tested C# solution

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Working console programs; Tested C# solution, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-011 | .NET Full Stack | NT-C-011-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	SQL, schema design and Entity Framework Core; HTML, CSS, JavaScript and responsive UI
Completion evidence	Persistent data model; Responsive front end
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 3 - Database layer

Objective: SQL, schema design and Entity Framework Core

Observable evidence: Persistent data model

Concept	Approved knowledge statement	Application behaviour	Evidence
SQL	A language for defining, querying and controlling relational data. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when sql is appropriate; apply it in a .NET Full Stack task; verify the result.	Persistent data model
schema design	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when schema design is appropriate; apply it in a .NET Full Stack task; verify the result.	Persistent data model
Entity Framework Core	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when entity framework core is appropriate; apply it in a .NET Full Stack task; verify the result.	Persistent data model

Module 4 - Web foundation

Objective: HTML, CSS, JavaScript and responsive UI

Observable evidence: Responsive front end

Concept	Approved knowledge statement	Application behaviour	Evidence
HTML	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when html is appropriate; apply it in a .NET Full Stack task; verify the result.	Responsive front end
CSS	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when css is appropriate; apply it in a .NET Full Stack task; verify the result.	Responsive front end
JavaScript	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when javascript is appropriate; apply it in a .NET Full Stack task; verify the result.	Responsive front end
responsive UI	A layout that adapts to different screen sizes and input methods. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when responsive ui is appropriate; apply it in a .NET Full Stack task; verify the result.	Responsive front end

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Persistent data model; Responsive front end, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-011 | .NET Full Stack | NT-C-011-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	MVC, routing, configuration and dependency injection; REST, validation, authentication and authorization
Completion evidence	Server-side application; Secured API
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 5 - ASP.NET Core

Objective: MVC, routing, configuration and dependency injection

Observable evidence: Server-side application

Concept	Approved knowledge statement	Application behaviour	Evidence
MVC	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when mvc is appropriate; apply it in a .NET Full Stack task; verify the result.	Server-side application
routing	Selection and forwarding of network paths between destinations. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when routing is appropriate; apply it in a .NET Full Stack task; verify the result.	Server-side application
configuration	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when configuration is appropriate; apply it in a .NET Full Stack task; verify the result.	Server-side application
dependency injection	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dependency injection is appropriate; apply it in a .NET Full Stack task; verify the result.	Server-side application

Module 6 - APIs and security

Objective: REST, validation, authentication and authorization

Observable evidence: Secured API

Concept	Approved knowledge statement	Application behaviour	Evidence
REST	An http-oriented interface style using resources, methods and predictable responses. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when rest is appropriate; apply it in a .NET Full Stack task; verify the result.	Secured API
validation	Checking input or output against defined rules before trusting it. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a .NET Full Stack task; verify the result.	Secured API
authentication	Verification of identity. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when authentication is appropriate; apply it in a .NET Full Stack task; verify the result.	Secured API
authorization	Control of what an authenticated identity is allowed to do. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when authorization is appropriate; apply it in a .NET Full Stack task; verify the result.	Secured API

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Server-side application; Secured API, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-011 | .NET Full Stack | NT-C-011-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	React or Angular, forms, state and API calls; Testing, logging, Git, deployment and documentation
Completion evidence	Integrated interface; Deployed capstone
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 7 - Front-end integration

Objective: React or Angular, forms, state and API calls

Observable evidence: Integrated interface

Concept	Approved knowledge statement	Application behaviour	Evidence
React	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when react is appropriate; apply it in a .NET Full Stack task; verify the result.	Integrated interface
Angular	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when angular is appropriate; apply it in a .NET Full Stack task; verify the result.	Integrated interface
forms	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when forms is appropriate; apply it in a .NET Full Stack task; verify the result.	Integrated interface
state	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when state is appropriate; apply it in a .NET Full Stack task; verify the result.	Integrated interface
API calls	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when api calls is appropriate; apply it in a .NET Full Stack task; verify the result.	Integrated interface

Module 8 - Quality and deployment

Objective: Testing, logging, Git, deployment and documentation

Observable evidence: Deployed capstone

Concept	Approved knowledge statement	Application behaviour	Evidence
Testing	Systematic verification that expected behaviour and important failure cases work correctly. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when testing is appropriate; apply it in a .NET Full Stack task; verify the result.	Deployed capstone
logging	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logging is appropriate; apply it in a .NET Full Stack task; verify the result.	Deployed capstone
Git	Distributed version control for tracking and reviewing source changes. In .NET Full Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when git is appropriate; apply it in a .NET Full Stack task; verify the result.	Deployed capstone
deployment	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployment is appropriate; apply it in a .NET Full Stack task; verify the result.	Deployed capstone
documentation	A defined competency within .NET Full Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a .NET Full Stack task; verify the result.	Deployed capstone

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Integrated interface; Deployed capstone, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-011 | .NET Full Stack | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-011-SK01	C# foundations	Prerequisite foundation	Working console programs	Critical
NT-C-011-SK02	Professional C#	M01 plus earlier foundations	Tested C# solution	Critical
NT-C-011-SK03	Database layer	M02 plus earlier foundations	Persistent data model	Supporting
NT-C-011-SK04	Web foundation	M03 plus earlier foundations	Responsive front end	Supporting
NT-C-011-SK05	ASP.NET Core	M04 plus earlier foundations	Server-side application	Supporting
NT-C-011-SK06	APIs and security	M05 plus earlier foundations	Secured API	Critical
NT-C-011-SK07	Front-end integration	M06 plus earlier foundations	Integrated interface	Supporting
NT-C-011-SK08	Quality and deployment	M07 plus earlier foundations	Deployed capstone	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-011 | .NET Full Stack | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-011-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of C# foundations and Professional C#?	The learner should provide Working console programs; Tested C# solution and explain how the work applies Syntax, control flow, methods and types; OOP, collections, LINQ, async and testing. Recall alone is insufficient.
NT-C-011-S01-EX02	Scenario/remediation	A learner reports completing Professional C# but cannot produce Tested C# solution. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-011-S01, request Tested C# solution, and allow a new equivalent test only after remediation.
NT-C-011-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Database layer and Web foundation?	The learner should provide Persistent data model; Responsive front end and explain how the work applies SQL, schema design and Entity Framework Core; HTML, CSS, JavaScript and responsive UI. Recall alone is insufficient.
NT-C-011-S02-EX02	Scenario/remediation	A learner reports completing Web foundation but cannot produce Responsive front end. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-011-S02, request Responsive front end, and allow a new equivalent test only after remediation.
NT-C-011-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of ASP.NET Core and APIs and security?	The learner should provide Server-side application; Secured API and explain how the work applies MVC, routing, configuration and dependency injection; REST, validation, authentication and authorization. Recall alone is insufficient.
NT-C-011-S03-EX02	Scenario/remediation	A learner reports completing APIs and security but cannot produce Secured API. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-011-S03, request Secured API, and allow a new equivalent test only after remediation.
NT-C-011-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Front-end integration and Quality and deployment?	The learner should provide Integrated interface; Deployed capstone and explain how the work applies React or Angular, forms, state and API calls; Testing, logging, Git, deployment and documentation. Recall alone is insufficient.
NT-C-011-S04-EX02	Scenario/remediation	A learner reports completing Quality and deployment but cannot produce Deployed capstone. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-011-S04, request Deployed capstone, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Programming/Platform blueprint. Human-review every scored item before initial production release.

NT-C-011 | .NET Full Stack | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Inventory or task manager	C#, SQL and CRUD	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Support-ticket API	ASP.NET Core, EF Core and validation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Admissions portal	Security, UI integration, testing and deployment	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.

NT-C-011 | .NET Full Stack | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior .NET Developer	C# 25% [critical]; OOP 25% [critical]; SQL 25%; Git 25%	Tested solution repository	If a critical skill is below 60 or tested solution repository is missing, return Improve First or Needs Evidence with the linked module/project.
2	ASP.NET Core Developer - Junior	MVC/API 34% [critical]; EF Core 33% [critical]; security 33%	Deployed API or web app	If a critical skill is below 60 or deployed api or web app is missing, return Improve First or Needs Evidence with the linked module/project.
3	Full Stack .NET Developer	Front end 34% [critical]; backend 33% [critical]; database 33%	End-to-end capstone	If a critical skill is below 60 or end-to-end capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-011 | .NET Full Stack | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is .NET Full Stack for?	Learners targeting Microsoft-stack web application roles.
What should I know before starting?	Basic logic and computer use; prior coding is helpful.
How long is the approved learning journey?	20-24 weeks / 240-300 guided hours.
What will I be able to demonstrate?	Build, test and deploy a secure database-backed application with ASP.NET Core APIs and a modern user interface.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
.NET Full Stack knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-011; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```